ITMO UNIVERSITY

LECTURE 1 · ITMO UNIVERSITY · 23 SEPTEMBER 2026

Introduction to Social Media Analytics

From posts and platforms to AI agents that read — and now write — the feed.

Prof. Aleks Farseev · Daniel Onishchenko · Jack Zhao

Lecturer · Tutors

1 The new information age

2 Platforms in 2026

3 Observatories → AI intelligence

4 Challenges

5 Course arrangements

YOUR FACILITATORS

Research, industry and teaching — in one room.



Prof. Aleks Farseev

COURSE LEAD

Founder & CEO, SOMIN.AI. PhD, National University of Singapore. 30+ papers at AAAI, ACM Multimedia, SIGIR, TOIS.



Daniel Onishchenko

FACILITATOR

DEFY Research. Lead author of EDRA, self-improving LLM outreach agents (ACM Multimedia 2026 demo).



Jack Zhao

FACILITATOR

ITMO University. Co-author of EDRA (ACM Multimedia 2026 demo).

LECTURE 1 · OUTLINE

Five questions for today.

01

The new information age

How big is it, and who — or what — is creating it?

02

Social media platforms in 2026

Who owns the feed, and how do they make money?

03

From observatories to AI intelligence

How we watch social data — then and now.

04

Challenges in social media

Users, content and models: what makes it hard.

05

Course arrangements

Grading, labs, the mini-project and the schedule.

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01

SECTION 1

The new information age

INTERNET TRAFFIC, 2024

“For the first time in a decade, bots — not humans — made up most of the internet's traffic: 51%.

Imperva's 2025 Bad Bot Report. Generative AI made automation cheap. Analysing social data now starts with asking who, or what, produced it.

THE NEW INFORMATION AGE · 2026 IN NUMBERS

Two-thirds of humanity is on social media. 1 in 8 uses AI.

6.04B

INTERNET USERS

73.2% of the world's 8.25 billion people are online.

5.66B

SOCIAL MEDIA IDENTITIES

68.7% of the population. About 21.5M new identities every month.

1B+

MONTHLY AI USERS

ChatGPT alone reports ~800M weekly users. Almost none of this existed in 2022.

$277B

SOCIAL MEDIA AD SPEND

2025, growing 13.6% a year. Digital is 74.4% of the $1.16T global ad market.

WHAT CHANGED SINCE THE LAST RUN OF THIS COURSE

- ✓ **Text → video → generated.** Most growth is in short video, and a rising share is made by AI.
- ✓ **Mobile and always on.** Users run ~6.75 platforms and spend 2.5+ hours a day on social and video.
- ✓ **Search moved into feeds and chatbots.** Discovery now happens in TikTok, Instagram and AI assistants.

SOURCE DataReportal / We Are Social / Meltwater, Digital 2026 Global Overview Report (Oct 2025); ad spend figures as cited there.

THE NEW INFORMATION AGE · FOUR MEDIA MODELS

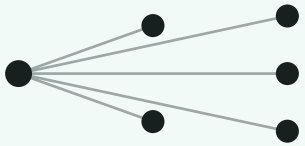
Social media publishes first and filters later. AI now generates first.



ONE → ONE

Communication media

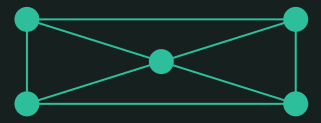
Phone, email, DMs



ONE → MANY

Broadcast media

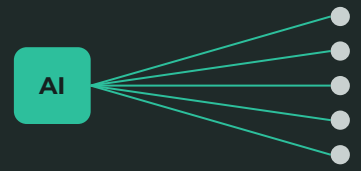
Filter, then publish



MANY → MANY

Social media

Publish, then filter



MACHINE → MANY

Generative media

Generate, then filter

BROADCAST MEDIA	Offers curated content. Editors control quality — or censor.
SOCIAL MEDIA	Offers no content of its own. Users create it; ranking algorithms decide who sees it.
GENERATIVE MEDIA	Content is produced on demand by models and agents. The scarce thing is trust, not content.

SOURCE Farseev, ITMO Social Media Analytics — Lecture 1 (media models); fourth model added for 2026.

THE NEW INFORMATION AGE · CHARACTERISTICS

Seven classic traits of social media — and the 2026 twist on each.

1 Everyone can be a media outlet

...and now every AI agent can be one too.

2 Rich user interactions

Likes, shares, comments, duets, remixes, DMs.

3 User-generated content

Shares with AI-generated content: 21% of Shorts to a new user were AI slop.

4 User-enriched content

Tags, comments and community notes add meaning.

5 Collaborative environment

Communities, groups, creator collabs.

6 Collective wisdom

Reviews, ratings — and the risk of fake consensus.

7 Long tail

Niche creators reach niche audiences via recommendation.

THE 2026 TWIST

21%

of the first 500 YouTube Shorts shown to a brand-new account were low-quality AI-generated "slop". A further 33% were "brainrot".

The top slop channel earns an estimated $4M+ a year. Generated content is already a business.

SOURCE Farseev, ITMO SMA Lecture 1 (characteristics); Kapwing, "AI Slop Report: The Global Rise of Low-Quality AI Videos" (Nov 2025, October 2025 data).

THE NEW INFORMATION AGE · STANFORD, SCIENCE (NOV 2025)

The algorithm is the editor. Stanford measured how much.

01

Intercept

A browser extension reads the X "For you" feed of each participant.

02

Classify

An LLM scores every post for anti-democratic attitudes and partisan animosity.

03

Rerank

Hostile posts are moved lower within seconds. Nothing is deleted.

04

Measure

Surveys before and after: how warm do users feel toward the other party?

WHY THIS MATTERS FOR ANALYTICS

Until now only platforms could run experiments on their own feeds. An LLM plus a browser extension let independent researchers do it — the same tools you will use in this course.

THE RESULT

~1,200

participants, 10 days, during the 2024 US election

2 pts

warmer toward the other party on a 100-point scale — about three years of natural attitude change.

SOURCE Piccardi, Saveski, Jia, Hancock, Tsai, Bernstein, "Reranking partisan animosity in algorithmic social media feeds alters affective polarization", Science (27 Nov 2025); Stanford Report.

02

SECTION 2

Social media platforms in 2026

META EARNINGS CALL, Q2 2026

> **Instagram: 2 billion daily actives. Threads: 500 million monthly — three years after launch.**

Meta's platforms reach 3.6 billion people every day. The growth now comes from AI recommendation, not from new users.

PLATFORMS · THE LANDSCAPE

Same categories as ten years ago. New winners, new users.

SOCIAL NETWORKS

Facebook, LinkedIn, VK

Private, professional and public networks around a profile.

SHORT VIDEO & MEDIA

YouTube, Instagram, TikTok

The fastest-growing category. Video-first, recommendation-first.

MESSAGING

WhatsApp, Telegram, WeChat, Max

Private by default. Channels and bots blur it with broadcast.

PUBLIC CONVERSATION

X, Threads, Bluesky

Microblogs: open follow graphs, real-time, text-heavy.

VISUAL DISCOVERY

Pinterest, Xiaohongshu (RedNote)

Curation and shopping inspiration. Search inside the app.

LOCAL & LOCATION

Google Maps, Yelp, Swarm, 2GIS

Check-ins moved into reviews, maps and delivery apps.

SOCIAL COMMERCE

TikTok Shop, Instagram, Meituan

The feed is the store. Creators sell directly.

AGENT-NATIVE

Moltbook

Launched Jan 2026: a social network where the users are AI agents.

SOURCE Farseev, ITMO SMA Lecture 1 (platform categories), updated with 2026 platforms; DataReportal Digital 2026.

PLATFORMS · META, Q2 2026

In 2026, a social platform is an AI ranking and ads company.

3.60B

DAILY ACTIVE PEOPLE

Across Facebook, Instagram, WhatsApp, Messenger and Threads (June 2026, +3% YoY).

$60.8B

QUARTERLY REVENUE

+28% YoY. $59.4B of it is advertising. Ad impressions +14%, price per ad +12%.

$75B

ADVANTAGE+ RUN-RATE

Annual revenue run-rate of Meta's AI-automated ad suite.

9M

SMALL BUSINESSES

use at least one of Meta's generative-AI ad creative tools.

RECOMMENDATION

LLM-based ranking: over half of recommended Instagram Feed content is now less than a day old — twice the share of a year earlier. Facebook video time +9% YoY.

THE LESSON FOR THIS COURSE

Profiling (L2), networks (L3), ranking (L4–6), LLMs (L7) and agents (L8) are not separate topics here. They are one revenue line.

SOURCE Meta, "Meta Reports Second Quarter 2026 Results" (press release) and Q2 2026 earnings call transcript (Motley Fool).

PLATFORMS · WHO OWNS THE FEED?

The platform dilemma, 2026 edition: the algorithm is the asset.

X (FORMERLY TWITTER)

From town square to AI division

MAR 2025	xAI acquires X in a deal valuing X at $33B.
DEC 2025	EU fines X €120M for blocking researcher data access (DSA Art. 40).
FEB 2026	SpaceX acquires xAI. X and Grok become SpaceX's AI division.
TODAY	Grok is built into the feed; API access is paid; research access is contested.

TIKTOK US

The recommender becomes geopolitics

JAN 2026	TikTok USDS Joint Venture closes. 80.1% non-Chinese ownership.
15% EACH	Oracle, Silver Lake and MGX. ByteDance keeps 19.9%.
ORACLE	Hosts US user data and oversees retraining the recommendation algorithm.
LESSON	Governments now regulate ranking models, not only content.

SOURCE Wikipedia, xAI (company); TechPolicy.Press on the EU's €120M X decision (Dec 2025); Variety / Hypebeast on TikTok USDS JV terms (Dec 2025); CNBC (Feb 2026).

PLATFORMS · CASE: MOLTBOOK

A social network for AI agents — where humans just watch. Mostly.

28 JAN 2026

Launch

Reddit-style network where OpenClaw agents post, comment and vote.

DAYS LATER

1.4M agents

Claimed. One researcher registered 500,000 accounts with a single script.

31 JAN 2026

Security hole

An unsecured database let humans control and impersonate agents.

10 MAR 2026

Meta acquires

Moltbook joins Meta Superintelligence Labs. Terms undisclosed.

WHY IT MATTERS HERE

- ✓ **Who is a user?** Counting "agents" failed the same way counting bot accounts always has.
- ✓ **Viral ≠ real.** The famous "secret agent language" post was faked by humans (MIT Tech Review: "AI theater").
- ✓ **Agents are the next audience.** Brands will market to agents that shop and choose for people.

SOURCE Wikipedia, Moltbook; Forbes (31 Jan 2026); TechCrunch, "Meta acquired Moltbook..." (10 Mar 2026).

Five questions to ask about any social platform.

01	**How many active users?**	MAU and DAU are the key metrics — and the easiest to inflate. Ask how "active" is defined, and whether bots count.
02	**Who are they?**	Demographics decide who advertises. Instagram skews young; Facebook older; LinkedIn professional.
03	**Public or private?**	Public networks are easy to analyse but harder to monetise. Private messaging is the reverse.
04	**How does it make money?**	Ads, then commerce, creator subscriptions, and now AI features and data licensing.
05	**What is the risk?**	Spam, bots, misinformation, privacy, regulation — and now AI-generated content at scale.

SOURCE Farseev, ITMO SMA Lecture 1 — "Analysis of Social Media Networks", updated for 2026.

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03

SECTION 3

From social observatories to AI marketing intelligence

TIM BERNERS-LEE, WEAVING THE WEB, 1999

> **Processes in which the people do the creative work and the machine does the administration.**

Berners-Lee called these "social machines". In 2026 the machine increasingly does the creative work too.

OBSERVATORIES · THEN AND NOW

From watching the crowd to asking AI what the crowd wants.

1999

Social machines

Berners-Lee: people create, machines administer.

2011

Reading the Riots

The Guardian traces rumours in 2.6M riot tweets.

2010s

Zooniverse

Citizen science: humans label what machines can't.

2014+

NUS Live Web Observatory

Live city analytics: fusing multi-source data.

2019+

SOMIN

Farseev's team turns social data into marketing insight.

2025–26

Agentic intelligence

Gemini-powered agents plan, create and score campaigns.

NUS Live Web Observatory research agenda: intelligent data gathering · deep NLP (English & Chinese) · event detection & tracking · predictive analytics · user and community profiles · from data to actions. Every item is a lecture in this course.

SOURCE Farseev, ITMO SMA Lecture 1 (social machines, observatories); The Guardian, Reading the Riots (2011); zooniverse.org; somin.ai.

OBSERVATORIES · MULTI-CHANNEL ANALYTICS

No source sees the whole user. Analytics starts with fusion.

SOCIAL MEDIA

Spontaneous user-generated content: posts, comments, video, check-ins.

WEB SOURCES

E-commerce sites, search engines, forums, reviews, news.

AD LIBRARIES

Meta Ad Library, Google Ads Transparency, TikTok Creative Center: what competitors run.

SENSORS & DEVICES

IoT and device-generated content: location, wearables, cars.

OFFLINE & PROPRIETARY

CRM, sales, surveys, loyalty data, domain and industry data.

GENERATED CONTENT

AI-written posts, synthetic reviews, agent traffic — to detect, filter or label.

THE FOURTH PARADIGM

Science was experimental, then theoretical, then computational. The fourth paradigm is data-intensive: hypotheses come from looking at data.

2026 twist: LLMs now propose the hypotheses too. Your job is to check them against the data.

SOURCE Farseev, ITMO SMA Lecture 1 — "Towards Multi-Channel Analytics"; J. Gray, The Fourth Paradigm (Microsoft Research, 2009).

OBSERVATORIES · CASE: SOMIN.AI (2026)

From public ad data to scored campaigns in 13 minutes.

INPUT

Public ad inventories from Meta, Google, TikTok and YouTube, plus social posts.

→

01

Brand Tracker

Decodes competitors' ads across channels by emotion, proposition and predicted performance.

→

02

Perspective Studies

Maps audience tensions and finds white-space opportunities in a category.

→

03

Expressions

Generates briefs and creative concepts, pre-tagged by channel and persona.

→

04

Ad Scoring

Predicts click-through and engagement before any media is bought.

UNDER THE HOOD

- ✓ **Gemini 2.5 Pro** on Google Cloud for ad understanding and insight generation
- ✓ **SoCraft CTR models** 78% prediction accuracy (Google Cloud case study)
- ✓ **Six AI agents** strategy, influencer, social, paid media, CRM, reporting

SOURCE somin.ai (accessed Sep 2026); Google Cloud customer story, cloud.google.com/customers/somin-ai.

OBSERVATORIES · SOMIN IN PRACTICE

What AI-driven social intelligence changes for real teams.

SAMY ALLIANCE

12×

faster research; 78% CTR prediction accuracy. Helped win a multi-million-pound UK account.

MOTHERCARE SINGAPORE

74%

research cost reduction; 25+ hours a week saved; 3× more granular audience segments.

KPI MEDIA

1 wk → 2 d

pitch cycle; +$5K monthly retainer growth.

CLICK2VIEW

9×

faster reporting; analysis 6× faster.

READ IT LIKE AN ANALYST

Vendor-reported case studies measure speed and cost well, and business impact less well. Ask: compared to what baseline? Over how many campaigns? Who measured it?

SOURCE somin.ai case studies (accessed Sep 2026); Google Cloud customer story on SOMIN (SAMY figures).

OBSERVATORIES · RESEARCH FROM THIS TEAM, 2025–2026

The frontier: explainable, agentic — and resistant to sameness.

ACM SIGIR 2026

SOMIN: Agentic AI for Automating Professional Visibility and Countering Content Homogenization

Agents that keep a professional visible online — without making everyone sound the same.

ACM MULTIMEDIA 2026 · DEMO

EDRA: self-improving LLM outreach agents

Lead author Daniel Onishchenko, with Jack Zhao. Agents that rewrite their own outreach rules. Case study in Lecture 8.

ACM MULTIMEDIA 2025

MindFuse: Towards GenAI Explainability in Marketing Strategy Co-Creation

Makes an LLM's strategy recommendations traceable to evidence.

WWW 2025 COMPANION

SOEX: Explainable AI for Competitor Communications Analytics

Explains what competitors say, and why it works.

ON STAGE & IN PRINT

ACM MM 2025 industry talk

Will AI make agencies obsolete? Replacement, Empowerment, Insight.

Forbes, Jul 2025

18 Powerful Applications of AI Agents in Marketing.

ATxSG 2026, Singapore

From 6 weeks to 13 minutes: pitching SOMIN AI.

SOURCE ACM Digital Library (doi.org/10.1145/3805712.3808371); farseev.com; acmmm2025.org/industry-expert-talks; forbes.com/councils/forbestechcouncil/people/aleksfarseev.

04

SECTION 4

Challenges in social media analytics

THE DATA IS DISAPPEARING

> **Meta shut down CrowdTangle on 14 August 2024. X charges for its API. Researchers now go to court for data.**

The data got bigger, noisier and harder to reach — all at once. That is the challenge this course prepares you for.

CHALLENGES · RECAP

Today's social data is live, big and 3M — plus a fourth M.

MULTI-SOURCE

Posts, search, e-commerce, ad libraries, sensors, CRM.

MULTIMEDIA

Short video and images first; text is the minority.

MULTILINGUAL

Informal, abbreviated, code-switched, fast-changing.

MACHINE-MADE

A growing share is written, filmed or posted by AI.

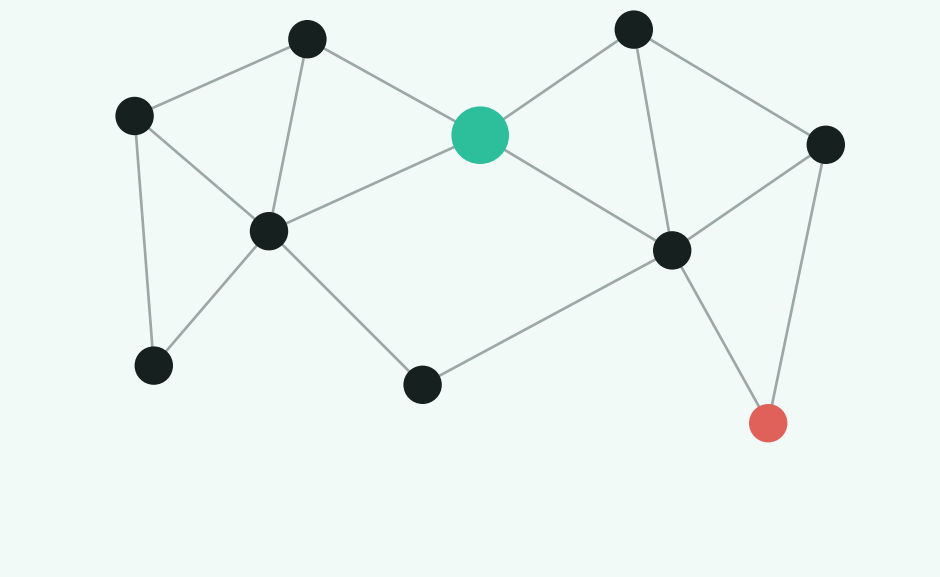
REQUIREMENTS FOR UNSTRUCTURED DATA ANALYTICS

- ✓ **Handle unstructured, multimodal, multi-source data** at scale and live.
- ✓ **Handle noise, misinformation and privacy** — now including synthetic content.
- ✓ **Offer insights** about content, knowledge, users and use cases.
- ✓ **Generate value for users and communities** — beyond advertising and e-commerce.

SOURCE Farseev, ITMO SMA Lecture 1 — "Key characteristics of current social media systems" and "Requirements for unstructured data analytics".

CHALLENGES · USER LEVEL

Users are a network. Networks can be measured — and faked.



Vertices = users. **Edges** = relationships, explicit (follow, friend) or implicit (similar behaviour). Stored as an adjacency matrix or list.

- ✓ **Find influential users.** Teal node: high centrality. Followers ≠ influence (Lecture 3).
- ✓ **Detect communities.** Dense clusters, across different domains and platforms.
- ✓ **Infer preferences and mobility.** What users like, where they go, what they will do next.
- ✓ **Profile users.** Age, gender, personality, interests — individual and community level (Lecture 2).
- ✓ **Spot fake accounts and agents.** Red node: behaves like a bot. Moltbook made this urgent.

SOURCE Farseev, ITMO SMA Lecture 1 — "Social network" and user-level challenges; Lectures 2–3 of this course.

CHALLENGES · CONTENT LEVEL

Content is visual, live, noisy — and increasingly synthetic.

01

Image & video

Instagram, YouTube and TikTok are video-first. Text-only analytics misses most of the signal. Multimodal models now read video directly.

02

Live location

Phones and sensors everywhere. Location traces, points of interest and audio feed mobility analysis and local recommendation.

03

Quality of data

The old lecture: >30% of UGC is noise, spam or rumours. In 2024, 51% of web traffic was automated and 37% was bad bots.

04

Structure

Most useful data is unstructured. LLMs now extract structure — entities, sentiment, claims — cheaply, but they can invent it too.

05

Scale

Big, dynamic, streaming data. Analysis must be incremental, cheap per item, and robust to drift.

SOURCE Farseev, ITMO SMA Lecture 1 — content-level challenges; Imperva (Thales), 2025 Bad Bot Report.

CHALLENGES · MODEL LEVEL

From passive observation to active AI that acts on the network.

Online + offline integration

Online data covers some segments of the population; offline data covers others. You need both for the full picture.

Privacy vs transparency

Users want privacy; society wants platforms audited. CrowdTangle closed; the EU's DSA forces researcher access.

Content homogenization

When everyone writes with the same LLM, everything sounds the same — for people and for brands.

New AI paradigms

From passive social networks to active ones: LLMs and agents that co-create, reply and act with users.

THE ROADMAP

L2–3	**Who is the user?**
L4–6	**What should they see?**
L7–8	**LLMs and agents that act**
L9–10	**Marketing and business value**

SOURCE Farseev, ITMO SMA Lecture 1 — model-level challenges; SOMIN, ACM SIGIR 2026 (content homogenization).

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SECTION 5

Course arrangements

THE DEAL

> **Two labs to learn the tools. One mini start-up to prove it. A defence in January.**

Everything you need to plan the semester: aims, grading, labs, the project and every deadline.

What this course covers — and what you will be able to do.

AIMS

The technical side of social media analytics, LLMs and agents, applied to marketing and business:

- ✓ **Content:** what users share and how it spreads
- ✓ **Relationships:** the networks between users
- ✓ **Analytics:** profiling, retrieval, recommendation, metrics
- ✓ **Generation & action:** LLMs and agents on top of it all

BY THE END, YOU SHOULD BE ABLE TO

- ✓ **Design, build and deploy** a working social-media or AI product with a real use case.
- ✓ **Explain** how the systems behind feeds, ads and agents work — and where they fail.
- ✓ **Evaluate** with the right metric, not just accuracy.
- ✓ **Judge** the implications for users, businesses and society.

SOURCE Farseev, ITMO SMA Lecture 1 — aims, objectives and teaching outcomes, extended for LLMs and agents.

COURSE ARRANGEMENTS · ASSESSMENT

Grades are earned by doing the work — on time.

20%

ATTENDANCE

Lectures and discussions. Showing up and taking part is part of the grade.

30%

TWO LABS

Individual. Points for quality — and for handing in on time.

50%

PROJECT + DEFENCE

Teams of 2–3. Judged on the product you build and how you defend it.

WHAT THE PROJECT IS JUDGED ON

- **Idea & design**
- **Technology**
- **Implementation, demo & report**
- **Going the extra mile**

THE RULES

- ✓ **Late = lost points.** Every late submission costs marks.
- ✓ **Knowing isn't enough.** No coursework barely earns a C.
- ✓ **Relative grading.** Letter grades follow the NUS system.
- ✓ **No copy-paste.** We will ask how your code works.

SOURCE Course teaser (16 Sep 2026) — grading split and rules; project criteria from the previous course's rubric.

COURSE ARRANGEMENTS · LABS AND MINI-PROJECT

Hands-on from week two.

LAB 1 · INDIVIDUAL

Engagement prediction

151k real professional posts from our EDRA research. For each post, predict whether it beat what its own author usually gets — below, typical or above. Full brief in the appendix.

LAB 2 · INDIVIDUAL

Multi-source prediction

A harder task fusing text, images and behaviour — e.g. predicting user demographics or personality, or whether a post will go viral.

MINI-PROJECT · TEAMS OF 2–3

A mini start-up: design and build a social-media or AI app with a strong use case.

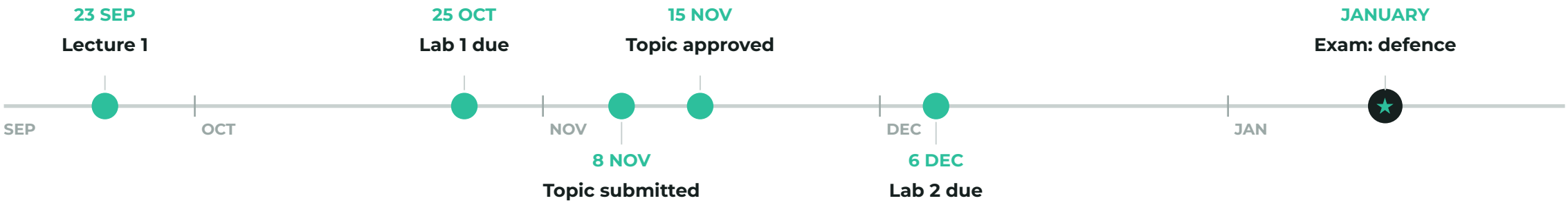
- ✓ **Novel, useful, compelling.** A real problem, a real user.
- ✓ **Uses course methods:** profiling, networks, retrieval, RecSys, LLMs, agents.
- ✓ **Topic approved first,** then checkpoints with the tutors.
- ✓ **Defended live in January:** demo, results, technical Q&A.

SOURCE Farseev, ITMO SMA Lecture 1 — labs and mini-project; course teaser (16 Sep 2026). Exact lab tasks announced with each release.

COURSE ARRANGEMENTS · SCHEDULE

Every deadline, from today to the defence.

LECTURES 1–10 weekly, from 23 Sep



LAB 1

Sun 25 Oct 2026, 23:59

Released after Lecture 2. Individual; predictions + notebook.

MINI-PROJECT TOPIC

Sun 8 Nov 2026, 23:59

One page per team: problem, user, data, methods.

TOPIC APPROVAL

By Sun 15 Nov 2026

Feedback from the tutors; revise and resubmit if asked.

LAB 2

Sun 6 Dec 2026, 23:59

Individual; multi-source prediction task.

EXAM

January 2027

Demo, results, technical Q&A. Exact day and slots announced in December.

SOURCE Deadlines are 23:59 Moscow time unless announced otherwise. Late submissions lose points. Updates are posted on the course Telegram channel.

COURSE ARRANGEMENTS · GO DEEPER

Recent lectures and reading from Stanford, MIT and the field.

01	NETWORKS	**Stanford CS224W —** J. Leskovec — Machine Learning with Graphs, Lecture 1: Introduction web.stanford.edu/class/cs224w/slides/01-intro.pdf
02	SOCIAL DATA + NLP	**Stanford CS224C —** D. Yang — NLP for Computational Social Science web.stanford.edu/class/cs224c
03	DEEP LEARNING	**MIT 6.S191 (2026) —** Introduction to Deep Learning — lectures Mar–May 2026 introtodeeplearning.com
04	STATE OF AI	**Stanford HAI —** The 2026 AI Index Report hai.stanford.edu/ai-index/2026-ai-index-report
05	FEED ALGORITHMS	**Science, Nov 2025 —** Piccardi et al. (Stanford) — Reranking partisan animosity in social media feeds science.org/doi/10.1126/science.adu5584
06	SOCIAL MEDIA STATS	**DataReportal —** Digital 2026 Global Overview Report datareportal.com/reports/digital-2026-global-overview-report

OTHER DETAILS · NEXT LESSON

See you next week: Lecture 2 — Social media user profiling.

- **Announcements:** Posted regularly on the course Telegram channel.
- **Lecture notes:** Posted for download before each lecture.
- **References:** Mainly from the web; 1–2 references per lecture, plus the Go deeper list.
- **Office hours:** Prof. Aleks Farseev: 12 PM on Zoom, pre-arranged via Daniel Onishchenko or Jack Zhao.
- **Labs & project:** Questions go to the tutors, Daniel Onishchenko and Jack Zhao.

BEFORE LECTURE 2

Look at the "Why am I seeing this ad?" page on Instagram or Facebook. What does the platform think it knows about you?

Next week: how a platform infers age, gender, personality and interests from what you post, where you go and what you look at — and where profiling goes too far.

A

APPENDIX

Lab 1: the dataset

LAB 1 · DUE SUN 25 OCT 2026, 23:59

> **Did this post beat what its own author usually gets? Below, typical or above.**

151,281 real posts from 5,439 professionals, collected for the EDRA research. Individual. Released after Lecture 2.

APPENDIX · LAB 1 · THE DATA

Not a benchmark. Real outreach data, collected for this research.

Every post was published by a real professional and crawled for the EDRA study. You get the same corpus the research team works with — the raw provider responses, nothing pre-digested.

151,281

POSTS

Text, images, video and slide documents, with engagement counts and a publication date on every one.

5,439

AUTHORS

A median of 24 posts each, plus a profile with career, skills and — for 98.7% — a follower count.

55,506

POSTS WITH MEDIA *

42,700 with images, 10,787 with video, 2,019 slide decks with extracted page text.

Fully pseudonymised. Names, handles, URLs, post IDs and company mentions become stable codes — the same organisation carries the same code in a post and in a career history, so trajectories stay learnable. Profile photographs are not released.

* Media type and count are released for every post. Which image files ship depends on the anonymisation review still in progress.

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APPENDIX · LAB 1 · WHAT YOU GET

Raw JSON, split for you. The feature engineering is the lab.

```
lab1/
├── train/
│   ├── authors/AUTHOR_0001.json          # profile
│   └── posts/AUTHOR_0001.json            # posts + label
├── test/
│   ├── authors/AUTHOR_4xxx.json
│   ├── posts_history/….json              # labels kept
│   └── posts_target/….json               # labels removed
├── media/ASSET_ID.{jpg,mp4,pdf}  *
├── label.py                              # level + cutoffs
├── baseline.py
└── README.md
```

- ✓ **No feature table.** You get the provider's own JSON. Deciding what is a feature is the work.
- ✓ **Split by author, 80 / 20 —** 3,584 authors to train on, 896 held out. A person has 24 posts, so a row-level split tests on the same human it trained on.
- ✓ **The level function ships too.** label.py — the definition behind the label, and the feature you need on held-out authors.
- ✓ **Follower counts** for 5,372 authors, each with the moment it was read.
- ✓ **Media type and count** for every post, plus the text of 1,298 slide decks. *
- ✓ **A baseline you must beat,** scored on this same split.

RICHER THAN IT LOOKS Fields no summary table would keep: a **seven-way reaction breakdown**, mentions, company tags, reshares.

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APPENDIX · LAB 1 · WHAT YOU PREDICT

Did this post beat what its own author usually gets?

One label per post, three classes. Not the raw count: the count mostly tells you who posted. The label tells you whether the post worked.

THE SAME 100 REACTIONS, TWO AUTHORS

Author A — usually **17** per post	**×5.9** → **ABOVE**
Author B — usually **4,963** per post	**×0.02** → **BELOW**

Both authors are real, both drawn from the released corpus. An absolute target would let a model score by recognising B and never reading a word.

CLASS	THE POST DREW	SHARE
below	under ×0.86 of the author's usual	**33.5 %**
typical	between ×0.86 and ×2.13	**33.3 %**
above	over ×2.13 of the author's usual	**33.2 %**

LABELLED **126,295** of 151,281 posts. The rest are each author's first five, which have no history to be measured against.

"Usual" is the median of the author's strictly earlier posts, at least five of them, on log1p(reactions). Earlier only — a level computed over all of an author's posts would contain the very post it is used to judge. **Reactions** means all seven counters: plain likes (82.5% of them) plus Love, Celebrate, Support, Insightful, Funny and Curious.

APPENDIX · LAB 1 · HOW YOU ARE SCORED

One number: macro-F1 over the three classes.

PRIMARY METRIC

macro-F1 — the unweighted mean of the three per-class F1 scores. One number, and it cannot be won by collapsing everything onto a single class.

SECONDARY

Quadratic weighted kappa

The classes are ordered. Calling a post above when it was below must cost more than calling it typical, and macro-F1 charges both the same.

REPORTED

Accuracy, and the confusion matrix

Do not also send micro-F1: with one label per post it is identical to accuracy. Report accuracy, and never without the matrix beside it.

NOT ACCEPTED

Any score on your own split

Split by author or your number is not comparable to anyone else's. Carve validation out of the training authors only.

A reference baseline ships with the data — a few-shot LLM given the author's recent posts and asked for one of the three classes, scored on this exact split, its number in the README. Beating it is the bar; explaining why you beat it is the grade.

✓ **Post age is not content.** Engagement is a snapshot from crawl time, so older posts collected more. Say so if you use age.

✓ **Submit one class per post** in posts_target, plus the notebook. At the defence you explain why your features work.